

Assignment No.

04

Combinational Logic Design

FAROOQ ABDULLA

Online Internship on Analog and Digital VLSI Design Flow

Batch 01

Q1. Design and Simulate a 4-bit Up/Down Counter using Verilog HDL.

- a) Behavioral modeling.
- b) Synchronous reset implementation.
- c) Asynchronous reset implementation.
- d) Verify up-counting operation.
- e) Verify down-counting operation.

DESIGN CODE CODE:

UP DOWN COUNTER SYNCHRONOUS

```
module up_down_counter_sync(  
    input clk,  
    input rst,  
    input up_down,  
    output reg [3:0] count);  
  
    always @(posedge clk) begin  
        if (rst) begin  
            count <= 4'b0000;  
        end else begin  
            if (up_down)  
                count <= count + 1'b1;  
            else  
                count <= count - 1'b1;  
        end  
    end  
end  
  
endmodule
```

UP DOWN COUNTER ASYNCHRONOUS

```
module up_down_counter_async(  
    input clk,  
    input rst,  
    input up_down,  
    output reg [3:0] count);  
  
    always @(posedge clk or posedge rst) begin  
        if (rst) begin  
            count <= 4'b0000;  
        end else begin
```

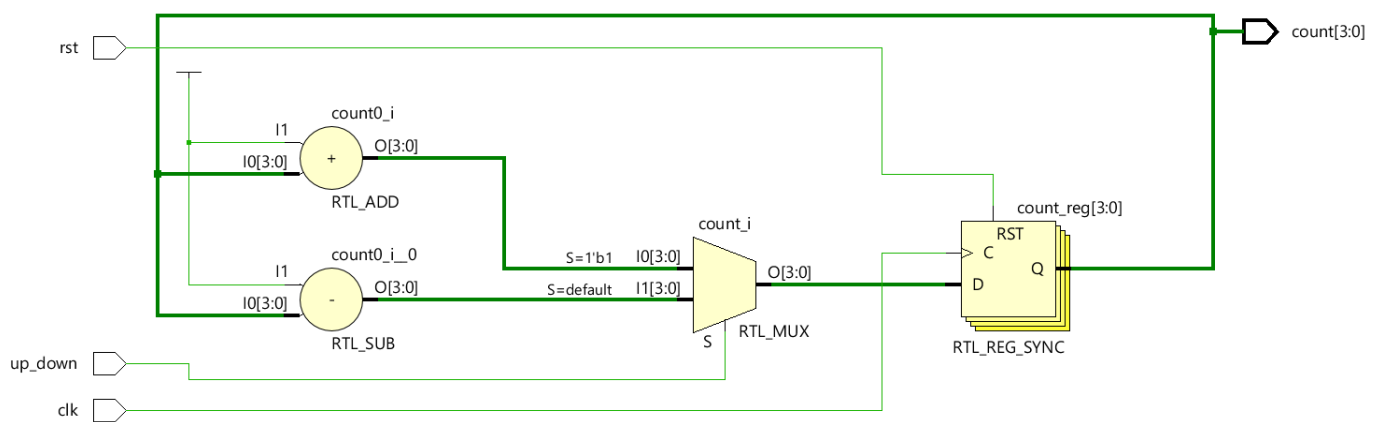
```

        if (up_down)
            count <= count + 1'b1;
        else
            count <= count - 1'b1;
        end
    end
endmodule

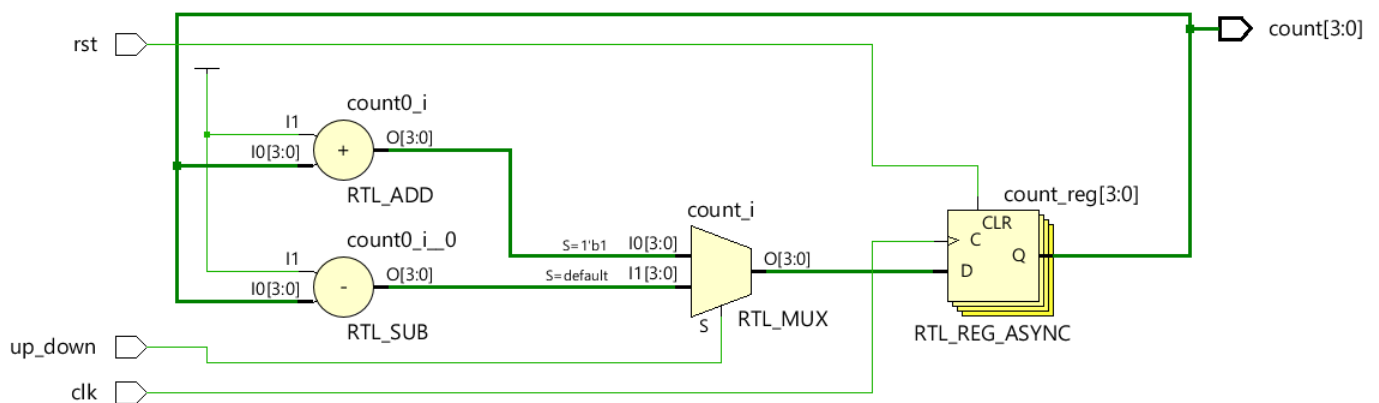
```

SCHEMATIC:

UP DOWN COUNTER SYNCHRONOUS



UP DOWN COUNTER ASYNCHRONOUS



TESTBENCH CODE:

UP DOWN COUNTER SYNCHRONOUS

```
module tb_up_down_counter_sync;

    reg clk;
    reg rst;
    reg up_down;

    wire [3:0] count_sync;

    up_down_counter_sync uut_sync ( .clk(clk), .rst(rst),
        .up_down(up_down), .count(count_sync));

    always #5 clk = ~clk;

    initial begin
        clk = 0;
        rst = 0;
        up_down = 1;

        $display("Time\t RST\t UP_DN\t Count_Sync\t ");
        $monitor("%0t\t %b\t %b\t %d\t\t", $time, rst, up_down, count_sync,);

        #2 rst = 1;
        #5 rst = 0;

        up_down = 1;
        #160;

        up_down = 0;
        #160;

        #3 rst = 1;
        #10;

        rst = 0;

        #20;
        $finish;
    end

endmodule
```

UP DOWN COUNTER ASYNCHRONOUS

```
module tb_up_down_counter_async;

    reg clk;
    reg rst;
    reg up_down;
    wire [3:0] count_async;

    up_down_counter_async uut_async (
        .clk(clk),
        .rst(rst),
        .up_down(up_down),
        .count(count_async));

    always #5 clk = ~clk;

    initial begin

        clk = 0;
        rst = 0;
        up_down = 1;

        $display("Time\t RST\t UP_DN\t Count_Sync\t Count_Async");
        $monitor("%0t\t %b\t %b\t %d", $time, rst, up_down, count_async);

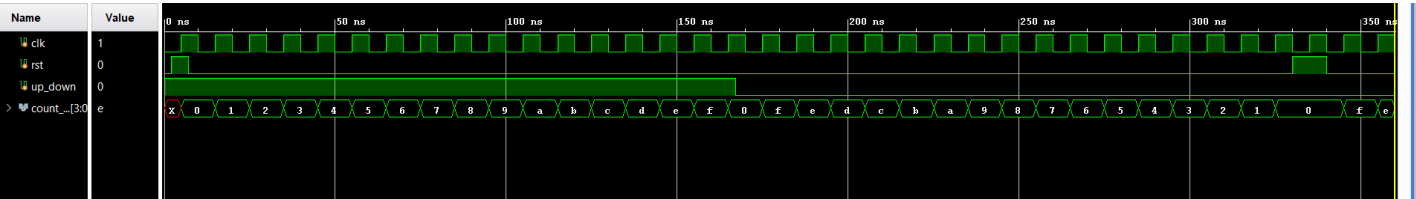
        #2 rst = 1;
        #5 rst = 0;

        up_down = 1;
        #160;
        up_down = 0;
        #160;

        #3 rst = 1;
        #10;
        rst = 0;

        #20;
        $finish;
    end
endmodule
```

SIMUTLATION:
UP DOWN COUNTER SYNCHRONOUS



UP DOWN COUNTER ASYNCHRONOUS

